25901226 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCECT202

Discipline Specific Elective (DSE) - MG4DSEECC200 - IOT SYSTEM DESIGN

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Which sensor is commonly used to measure tempreture? [U] / [CO1]

   a). Thermistor b). LDR c). PIR d). Ultrasonic

2. Which of the following is an example of an IoT gadget? [K] / [CO1]

   a). Smart watch b). Calculator c). Key board d). Printer

3. Smart energy meters used in factories are an example of: [U] / [CO3]

   a). Consumer IoT b). Industrial IoT

   c). Personal IoT d). Social IoT

4. Select the IoT device used for monitoring heart rate in healthcare applications. [K] / [CO3]

   a). Ultrasonic sensor b). Pulse sensor

   c). Gas sensor d). IR sensor

5. XMPP is primarily used for: [U] / [CO2]

a). File compression b). Database storage

c). Image processing d). Real-time messaging and presence

6. CoAP is mainly designed for: [U] / [CO2]

a). Cloud data centers b). Desktop applications

c). Multimedia streaming d). Constrained IoT devices

7. ESP8266 GPIO pins operate at [K] / [CO2]

a). 5 V logic level b). 3.3 V logic level

c). 12 V logic level d). Dual voltage

8. A gyroscope sensor is used to measure [K] / [CO2]

a). Linear speed b). Distance

c). Angular velocity d). Pressure

9. The GPIO pins of Raspberry Pi Zero operate at: [K] / [CO2]

a). 1.8V logic b). 3.3V logic

c). 5V logic d). 12V logic

10. Firebase Hosting is mainly used to: [K] / [CO2]

a). Train AI models b). Host web applications

c). Repair software d). Manage hardware drivers

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Define IoT.Explain main components of IoT system. [U] / [CO1]

12. Explain the working of IoT system. [U] / [CO1]

13. Explain privacy issues in IoT systems with simple examples. [U] / [CO3]

14. Explain Medical IoT (IoMT) and its importance in healthcare systems [K] / [CO3]

15. Explain the working principle of magnetic actuators with suitable examples. [U] / [CO2]

16. Explain the role of IR and PIR sensors in home automation systems with suitable examples. [A] / [CO2]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Explain the basic components of an embeded system with neat digram. [U] / [CO1]

18. Explain what are the changes needed to convert manual industrial monitoring into IoT-based monitoring. [A] / [CO3]

19. Describe the features of the ESP32 that make it suitable for IoT applications. Explain its power efficiency, connectivity options, and processing performance. [U] / [CO2]

20. Describe in detail the working of an ultrasonic sensor. Explain the distance measurement technique and its applications in IoT systems. [An] / [CO2]

[10x2 = 20]